

Adarsh Anand

Full Stack Engineer — AI Systems & Agentic Pipelines — Automation
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PROFESSIONAL SUMMARY

Full-Stack AI Engineer with 5+ years of experience designing and shipping production systems across the complete stack — from Next.js/React frontends to FastAPI/Node.js backends, real-time infrastructure, and LLM-powered applications. Specialises in multi-agent orchestration, RAG pipelines, and agentic system design, with hands-on delivery of systems integrating multi-LLM providers (Claude, GPT-4o, Gemini, Ollama), vector search (pgvector), and streaming AI interfaces. Proven track record building secure, observable, and scalable architectures — including end-to-end encrypted messaging (Signal Protocol), ETL pipelines with NLP/NER, and real-time WebSocket systems deployed across government-scale multilingual AI platforms. Brings strong engineering discipline: TypeScript-first frontend, Python for AI/data layers, and deep experience with async job queues, offline-first patterns, and developer tooling (VS Code Extension API).

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Frontend: React (18/19), Next.js (App Router), Tailwind CSS, Zustand, TanStack React Query, Vite

Backend: Node.js, Express.js, FastAPI, NestJS, Socket.IO, BullMQ

AI & Agentic: Multi-Agent Systems, RAG Pipelines, LLM Integration (Claude, GPT-4o, Gemini, Ollama), pgvector, Prompt Engineering

Databases: PostgreSQL, Redis, MongoDB, Supabase, Drizzle ORM, SQLAlchemy

Infrastructure: Docker, GitHub Actions, AWS · GCP · Azure, Nginx, PM2, Kubernetes

Testing & Tooling: Playwright, Vitest, Jest, pytest, Prometheus, Stripe, VS Code Extension API, No-Code Integration (N8N, Zapier), API Automation

EDUCATION

REVA University, Bangalore

(Aug 2017 - Sep 2021)

Bachelor of Technology, Computer Science (8 CGPA)

EXPERIENCE

Digital India – Bhashini (Ministry of Electronics & IT), Delhi (Aug 2025 – Present)
Senior AI - Full Stack Engineer

- Architected and led the development of a **multi-agent AI research platform** integrating intent detection, query planning, web search, scraping, knowledge-base construction, synthesis, and critic-based validation pipelines to produce citation-backed research reports.
- Engineered and scaled key components of the **BHASHINI ecosystem**, driving API reliability, platform observability dashboard, performance engineering, developer tooling, and architecture-led product innovation across multilingual AI services supporting **22+ Indian languages** and integrated with 3 ASR/TTS/NMT provider APIs.

AppSnare, Kanpur

(Dec 2023 – Jul 2025)

Full Stack Developer

- Architected full-stack AI applications using NestJS, FastAPI, PostgreSQL, and TypeScript with automated CI/CD pipelines — delivering **LLM-powered** systems with **RAG**, **vector embeddings (pgvector)**, and **semantic retrieval** for intelligent search and automated content processing.
- Designed asynchronous **ETL** and workflow orchestration pipelines with multi-language document ingestion (OCR, NER, fuzzy matching), job queues, and real-time SSE-based progress streaming to support high-throughput AI data applications.

Spektra Systems, Bangalore

(Mar 2021 – Oct 2023)

Cloud Developer

- Architected and deployed production-grade **SaaS products** on Microsoft Azure Marketplace, overseeing full cloud lifecycle.
- Built reusable components and infrastructure modules to boost performance.
- Automated build and deployment workflows using Python, Shell, and Docker, reducing operational overhead.

PROJECTS

SnipChat — Real-time E2E Encrypted collaboration ecosystem

Jan 2026

- Built a VS Code extension with end-to-end encrypted real-time messaging using Signal Protocol (X3DH + Double Ratchet), AES-256-GCM encryption, and offline-first sync via IndexedDB/Dexie.js — backed by Socket.IO with Redis Adapter for horizontal scaling, BullMQ job queues, MinIO object storage, Drizzle ORM, Stripe billing, Prometheus metrics, and Playwright E2E tests.

ETL + Knowledge Graph Pipeline

Nov 2025

- Designed a document ingestion pipeline with GLiNER NER, SPO triple extraction, entity resolution, and pgvector RAG — converting unstructured text into a searchable knowledge graph with confidence-scored relations.

FinDiro — Corporate Finance Intelligence Tool

Sep 2024 – Present

- Built predictive financial insights using combined historical, live market, and strategic data models for decision support.

Volunteering & Internship

IEEE, Bangalore

(Aug 2018 - Dec 2019)

Web Developer

- Maintained IEEE Bangalore Section website and contributed to event portals and chapter initiatives.

CERTIFICATIONS & ACHIEVEMENTS

- Completed **Deep Learning Specialization** by Andrew Ng - Coursera
- Published **Research Paper** on Virtual Assistant: D.I.V.A.
- Open-source contributor — GitHub: @adarshrazor

- Launched No-Code AI Automation template store: @N8N Store
- Streamer — Content Creator Youtube: @RazorCloak